

Symmetric Group Practice 1

1. Let α be the 5-cycle $(1\ 2\ 3\ 4\ 5)$.

a) Write each distinct power of α in cycle notation.

$$\alpha^1 = (1\ 2\ 3\ 4\ 5) \quad \alpha^2 = (1\ 3\ 5\ 2\ 4) \quad \alpha^3 = (1\ 4\ 2\ 5\ 3) \quad \alpha^4 = (1\ 5\ 4\ 3\ 2) \\ \alpha^5 = (1)$$

b) How many distinct powers do you get for a 5-cycle? 5

c) What is the connection between the powers of α and modular arithmetic?

$$\alpha^i = \alpha^j \quad \text{iff} \quad i \bmod 5 = j \bmod 5 \quad i - j = 5k \quad \text{for some } k \in \mathbb{Z}.$$

d) What is α^{-1} ?

$$(1\ 5\ 4\ 3\ 2)$$

For what positive power n is $\alpha^{-1} = \alpha^n$?

$$\alpha^{-1} = \alpha^4$$

e) Which exponent of α would you use to represent the identity permutation?

0 is best since it matches up with the identity element in $\mathbb{Z}_5$.

Definition: If x is any element of a group G , the order of x is the smallest positive integer n such that x^n is the identity in the group.

2. What is the order of the permutation $(1\ 2)(3\ 4\ 5)$? 6

3. What is the order of the permutation $(1\ 2)(3\ 4\ 5\ 6)$? 4

4. What is the order of the permutation $(1\ 2)(3\ 4\ 5)(6\ 7\ 8\ 9)$? 12

5. Find two elements of order 15 in S_9 .

There are 24,192 different elements of order 15 in S_9
You must have a 3-cycle and a 5-cycle with no objects in common.
 $(1\ 2\ 3)(4\ 5\ 6\ 7\ 8)$ and $(1\ 3\ 9\ 6\ 5)(2\ 7\ 4)$ will work.

6. What is the largest possible order for an element in S_9 ? S_{10} ? S_{11} ?

S_9 largest order is 20 $(a b c d)(e f g h i)$

S_{10} largest order is 30 $(a b)(c d e)(f g h i j)$

S_{11} largest order is still 30 same as in S_{10}

7. Let $\alpha = (1\ 2\ 3\ 4)$ and $\beta = (1\ 2\ 3)$ be permutations in S_4 (so β leaves 4 fixed).

a) Write each permutation in array form.

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

b) Write the product $\alpha\beta$ in both array form and in cycle form.

$$\alpha\beta = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{bmatrix} = (1\ 3\ 2\ 4)$$

c) What is the order of α ?

4

What is the order of β ?

3

d) What is the order of the product $\alpha\beta$?

4

8. Fill in the Cayley table for S_3 from our notes in class.

$\circ$	ϵ	α	β	γ	δ	φ
ϵ	ϵ	α	β	γ	δ	φ
α	α	ϵ	δ	φ	β	γ
β	β	γ	ϵ	α	φ	δ
γ	γ	β	φ	δ	ϵ	α
δ	δ	φ	α	ϵ	γ	β
φ	φ	δ	γ	β	α	ϵ

$$\epsilon = (1)$$

$$\alpha = (2\ 3)$$

$$\beta = (1\ 2)$$

$$\gamma = (1\ 2\ 3)$$

$$\delta = (1\ 3\ 2)$$

$$\varphi = (1\ 3)$$